

Home Lab 02: Configuring Examples of Security Control Types

1. Lab Overview

In this lab, I configured and tested four major types of security controls: **preventive, detective, directive, and corrective controls**. The lab was completed in a Windows Server 2019 virtual environment using a machine named **PC10** and a domain server named **DC10**.

The purpose of this lab was to understand how different security controls are used to reduce risk, monitor activity, guide user behavior, and restore systems to a secure state. This lab also helped me connect CompTIA Security+ concepts to real-world Windows administration and IT support tasks.

2. Business Scenario

As a security team member at **Structureality Inc.**, I was responsible for helping improve the organization's security posture. The lab focused on configuring, using, and testing different security controls to understand how each control type protects business systems.

The work included testing access to a restricted shared folder, enabling auditing for file deletion, creating a login warning banner, and using PowerShell hashing to detect and correct unauthorized file changes.

3. Lab Objective

The objective of this lab was to:

- Understand the difference between preventive, detective, directive, and corrective controls.
- Configure a preventive control using file share permissions.
- Configure a detective control using Windows auditing and Event Viewer.
- Configure a directive control using a login warning banner.
- Configure a corrective control using PowerShell and file hashing.
- Test each control to confirm that it worked as expected.

4. Lab Environment

Item	Details
Organization	Structureality Inc.

Client VM	PC10
Server VM	DC10
Operating System	Windows Server 2019
User Account Tested	Sam
Admin Account Tested	Jaime
Shared Folder	\\10.1.16.1\TOOLS
Local Folder	C:\LABFILES
Tools Used	Server Manager, File Explorer, Event Viewer, Local Security Policy, PowerShell, Sysinternals NotMyFault

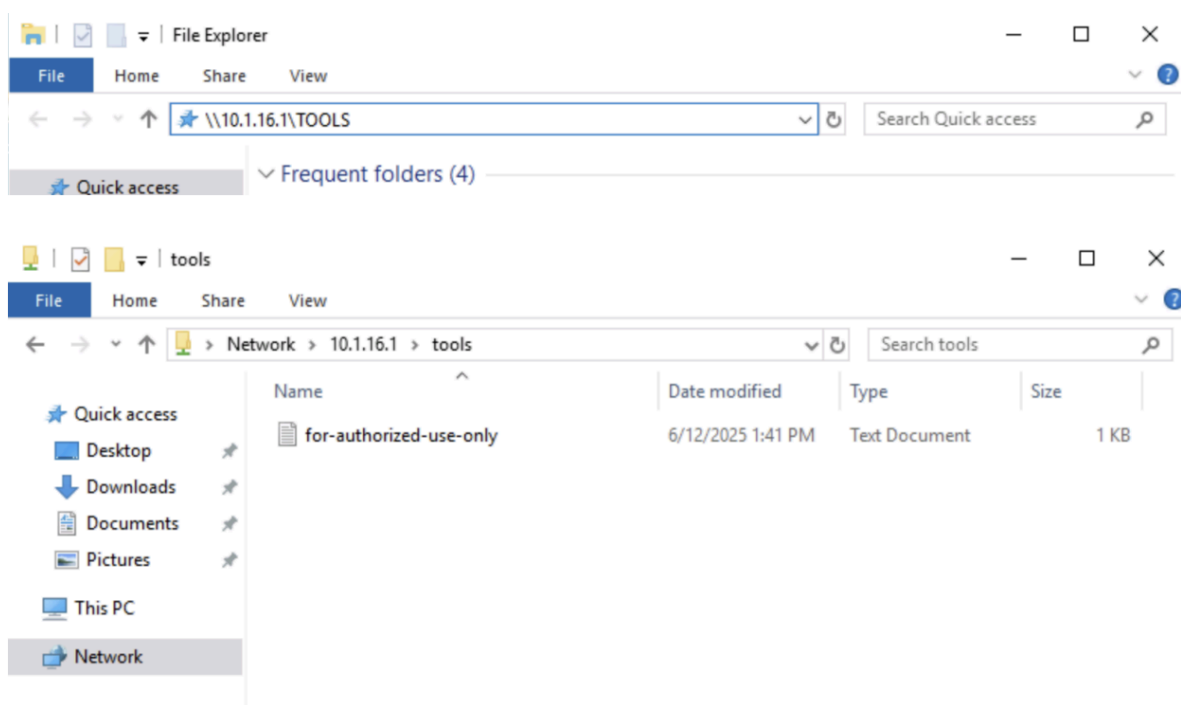


Figure 1: Non-administrative user Sam was initially able to access the TOOLS share.

5. Tools Used

- Windows Server 2019

- Server Manager
- File Explorer
- Event Viewer
- Local Security Policy
- PowerShell
- NTFS permissions
- Windows auditing
- Registry policy settings
- Sysinternals NotMyFault
- SHA256 file hashing

6. Control Types Covered

Control Type	Purpose	Lab Example
Preventive Control	Stops unwanted activity from succeeding	Restricted non-admin access to the TOOLS share
Detective Control	Records events and activities	Enabled auditing for folder deletion
Directive Control	Gives instructions to users	Created a login warning banner
Corrective Control	Restores a system to a preferred condition	Used PowerShell hash checking to restore file content

The primary purpose of preventive controls as stopping unwanted activity, detective controls as recording activity, directive controls as giving instructions, and corrective controls as restoring a system to a preferred condition.

7. Steps Performed

Part 1: Configured and Tested Preventive Controls

What I Did

The first part of the lab focused on a **preventive control**. A preventive control is designed to stop unwanted activity before it succeeds.

The shared folder `\\10.1.16.1\TOOLS` was supposed to be accessible only to administrators. I tested access using the non-administrative user **Sam**. Sam was able to view the share, which showed that the folder permissions were misconfigured.

Issue Found

The **Domain Users** group had inherited read permissions from the parent folder. This allowed non-administrative users like Sam to access a folder that should have been restricted.

Server Manager

File and Storage Services > Shares

Servers
Volumes
Disks
Storage Pools
Shares
iSCSI
Work Folders

SHARES
All shares | 3 total

Share	Local Path
DC10 (3)	
NETLOGON	C:\Windows\SYSVOL\sysvol\ad.str...
SYSVOL	C:\Windows\SYSVOL\sysvol
TOOLS	C:\TOOLS

VOLUME
TOOLS on DC10

(C:)
Capacity: 39.4 GB

33.2% Used
13.1 GB Used Space
26.3 GB Free Space

Go to Volumes Overview >

QUOTA
TOOLS on DC10

To use quotas, File Server Resource Manager must be installed.

To install File Server Resource Manager, start the Add Roles and Features Wizard.

TOOLS Properties

TOOLS

Show All

General

Permissions

Settings

General

Server Name: DC10

Share name: TOOLS

Share description: Shared TOOLS Folder

Folder path: C:\TOOLS

Protocol: SMB

Availability type: Not Clustered

TOOLS

- Show All
- General +
 - Permissions —
 - Settings +

Permissions

Permissions to access the files on a share are set using a combination of folder permissions, share permissions, and, optionally, a central access policy.

Share permissions: Everyone Full Control

Folder permissions:

Type	Principal	Access	Applies To
Allow	CREATOR OWNER	Full Control	Subfolders an
Allow	BUILTIN\Users	Special	This folder an
Allow	BUILTIN\Users	Read & execu...	This folder, su
Allow	BUILTIN\Administrators	Full Control	This folder, su
Allow	NT AUTHORITY\SYSTEM	Full Control	This folder, su
Allow	structureality\LocalAdmin	Read	This folder, su
Allow	structureality\Domain Admins	Full Control	This folder, su

< [Customize permissions...](#) >

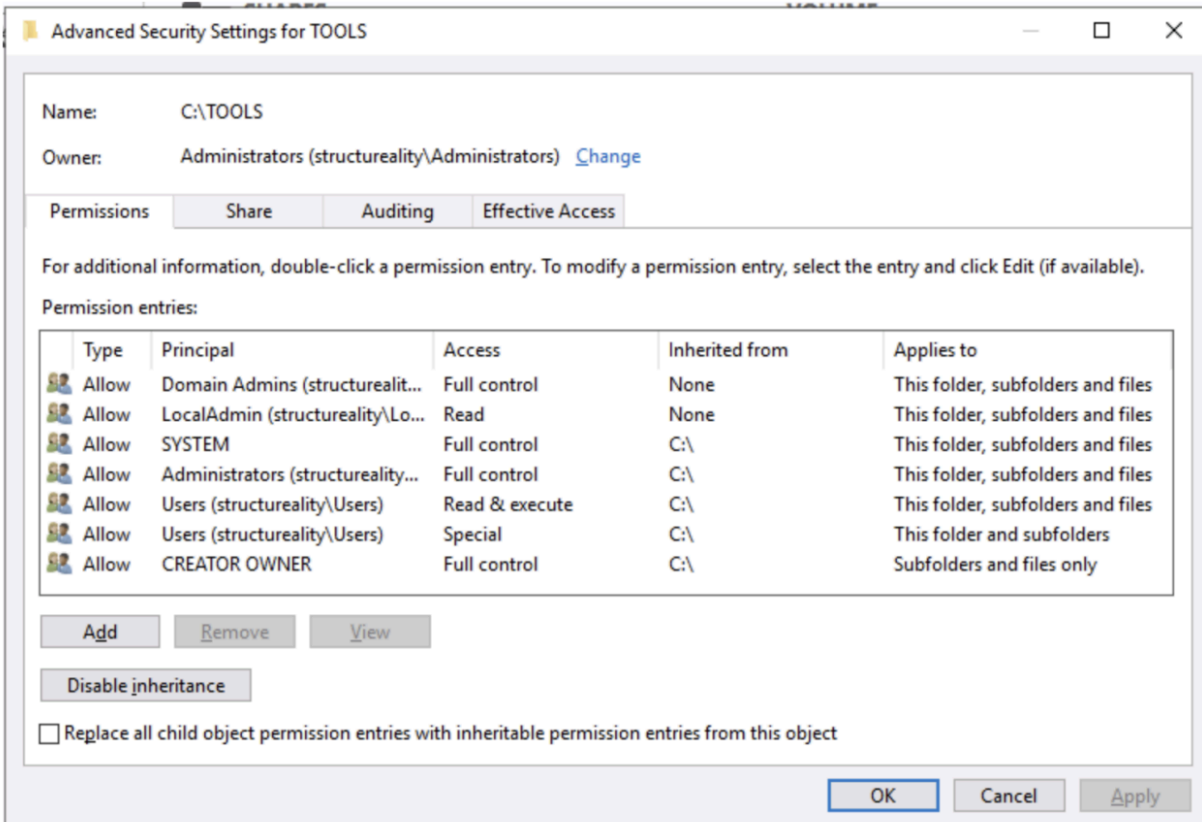


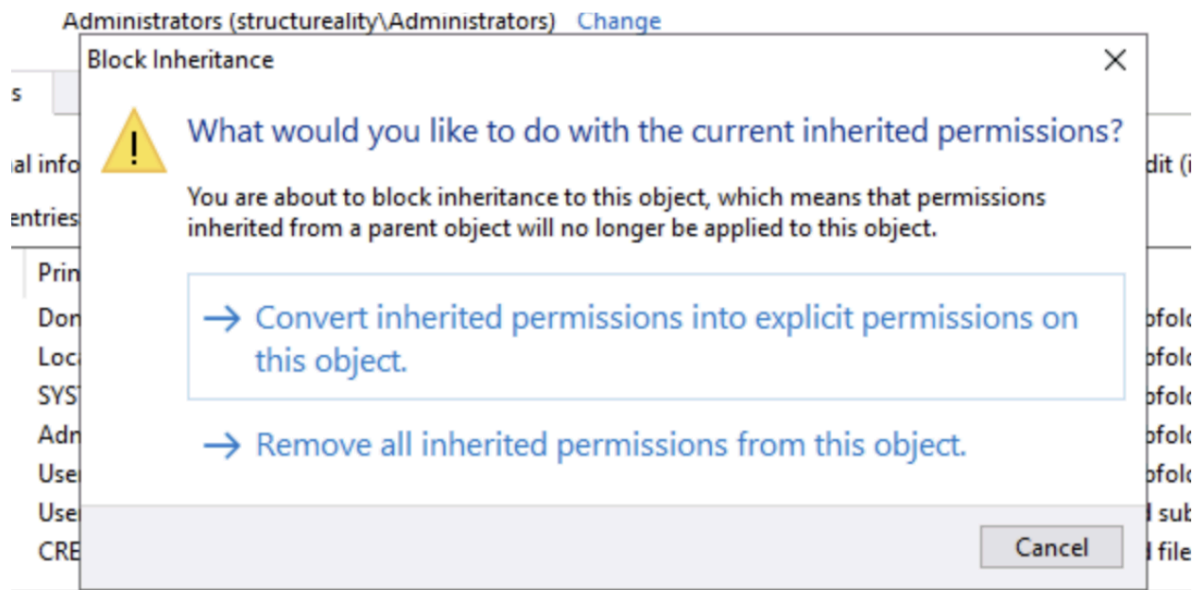
Figure 2: Inherited Users permissions allowed non-administrative access to the TOOLS share.

Action Taken

To fix the issue, I:

1. Logged in to DC10 as **Structureality\Administrator**.
2. Opened Server Manager.
3. Went to **File and Storage Services > Shares**.
4. Opened the properties for the **TOOLS** share.
5. Customized the permissions.
6. Disabled inheritance.
7. Converted inherited permissions into explicit permissions.
8. Removed the unnecessary **Users** permissions.
9. Removed the **Everyone** group from the ACL.
10. Tested access again using Sam.

After the change, Sam could no longer access the **TOOLS** share. This confirmed that the preventive control was working.



Permission entries:

Type	Principal	Access	Inherited from	Applies to
 Allow	Domain Admins (structurealit...	Full control	None	This folder, subfolders and files
 Allow	LocalAdmin (structureality\Lo...	Read	None	This folder, subfolders and files
 Allow	SYSTEM	Full control	None	This folder, subfolders and files
 Allow	Administrators (structureality...	Full control	None	This folder, subfolders and files
 Allow	Users (structureality\Users)	Read & execute	None	This folder, subfolders and files
 Allow	Users (structureality\Users)	Special	None	This folder and subfolders
 Allow	CREATOR OWNER	Full control	None	Subfolders and files only

[Add](#) [Remove](#) [Edit](#)

[Enable inheritance](#)

Permission entries:

Type	Principal	Access	Inherited from	Applies to
Allow	Domain Admins (structurealit...	Full control	None	This folder, subfolders and files
Allow	LocalAdmin (structureality\Lo...	Read	None	This folder, subfolders and files
Allow	SYSTEM	Full control	None	This folder, subfolders and files
Allow	Administrators (structureality...	Full control	None	This folder, subfolders and files
Allow	Users (structureality\Users)	Read & execute	None	This folder, subfolders and files
Allow	Users (structureality\Users)	Special	None	This folder and subfolders
Allow	CREATOR OWNER	Full control	None	Subfolders and files only

Add

Remove

Edit

Enable inheritance

TOOLS Properties

Advanced Security Settings for TOOLS

Name: C:\TOOLS

Owner: Administrators (structureality\Administrators) [Change](#)

Permissions Share Auditing Effective Access

For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).

Permission entries:

Type	Principal	Access	Inherited from	Applies to
Allow	Domain Admins (structurealit...	Full control	None	This folder, subfolders and files
Allow	LocalAdmin (structureality\Lo...	Read	None	This folder, subfolders and files
Allow	SYSTEM	Full control	None	This folder, subfolders and files
Allow	Administrators (structureality...	Full control	None	This folder, subfolders and files
Allow	CREATOR OWNER	Full control	None	Subfolders and files only

Add

Remove

Edit

Enable inheritance

☐ Replace all child object permission entries with inheritable permission entries from this object

OK

Cancel

Apply

Advanced Security Settings for TOOLS

Name: C:\TOOLS

Owner: Administrators (structureality\Administrators) [Change](#)

Permissions

Share

Auditing

Effective Access

To modify share permissions, select the entry and click Edit.

Network location for this share: \\DC10.ad.structureality.com\TOOLS

Permission entries:

	Type	Principal	Access
	Allow	Everyone	Full Control

Owner: Administrators (structureality\Administrators) [Change](#)

Permissions

Share

Auditing

Effective Access

To modify share permissions, select the entry and click Edit.

Network location for this share: \\DC10.ad.structureality.com\TOOLS

Permission entries:

No groups or users have permission to access this object. However, the owner of this object can assign permissions.

Name: C:\TOOLS
Owner: Administrators (structureality\Administrators) [Change](#)

Permissions **Share** Auditing Effective Access

For additional information, double-click a permission entry. To modify a permission entry, select the entry and click Edit (if available).

Permission entries:

	Type	Principal	Access	Inherited from	Applies to
	Allow	Domain Admins (structurealit...	Full control	None	This folder, subfolders and files
	Allow	LocalAdmin (structureality\Lo...	Read	None	This folder, subfolders and files
	Allow	SYSTEM	Full control	None	This folder, subfolders and files
	Allow	Administrators (structureality...	Full control	None	This folder, subfolders and files
	Allow	CREATOR OWNER	Full control	None	Subfolders and files only

Name: C:\TOOLS
Owner: Administrators (structureality\Administrators) [Change](#)

Permissions **Share** Auditing Effective Access

To modify share permissions, select the entry and click Edit.

Network location for this share: \\DC10.ad.structureality.com\TOOLS

Permission entries:

No groups or users have permission to access this object. However, the owner of this object can assign permissions.

Result

The preventive control successfully blocked non-administrative users from accessing administrator-only resources.



Sam

 →

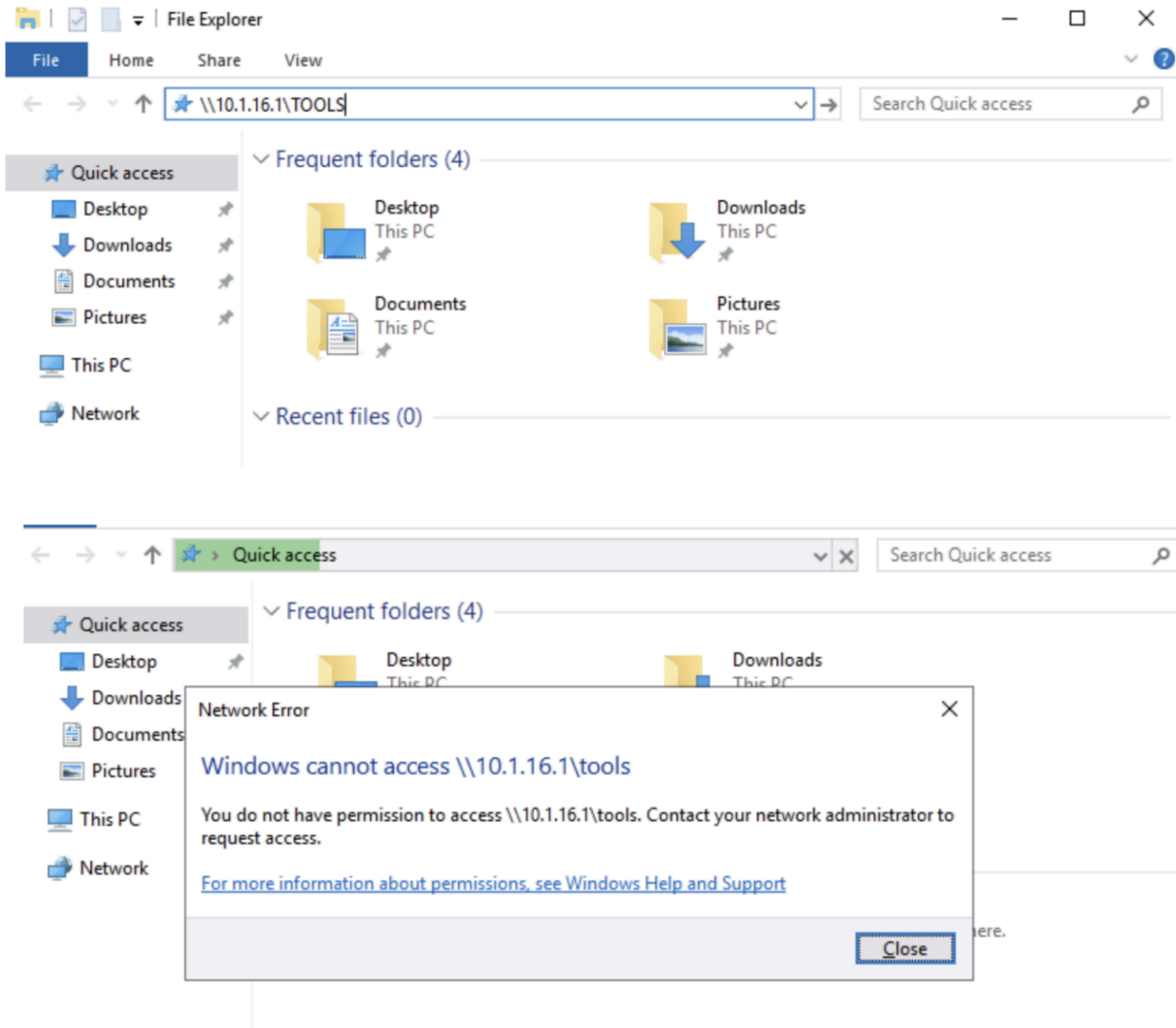


Figure 3: Sam was blocked from accessing the TOOLS share after the preventive control was applied.

Part 2: Configured and Tested Detective Controls

What I Did

The second part of the lab focused on a **detective control**. A detective control records events and activities so they can be reviewed later.

I first deleted a folder named **empty** from **C:\LABFILES**, then searched Event Viewer to see if the deletion was logged. The event was not found, which showed that folder deletion was not being audited.

LABFILES

Home Share View

This PC > Local Disk (C:) > LABFILES > Search LA

Name	Date modified	Type	Size
contains-nothing	10/22/2018 7:57 PM	File folder	
empty	10/22/2018 7:57 PM	File folder	
MARKETING	AM	File folder	
NVD-Control	AM	File folder	
pcaps	AM	File folder	
ScoutSuite	AM	File folder	
winlogbeat	AM	File folder	
515tech_store	3 AM	SQL File	8 KB
comptia-log	AM	JPG File	53 KB
CONFIDENTIAL	AM	Text Document	1 KB
conn-sample	AM	Text Document	155 KB
CSIRT Incident	PM	Microsoft Word D...	16 KB
CSIRT Incident	PM	Microsoft Edge P...	104 KB
DisableContr	PM	Windows PowerS...	1 KB
DisableDisk	0 AM	Windows PowerS...	1 KB
iam_shares	8 AM	Windows PowerS...	2 KB
laptop-full	AM	Wireshark capture...	321 KB
laptop-select	7 AM	Wireshark capture...	306 KB
local.rules	3/10/2020 5:26 AM	RULES File	1 KB

- Open
- Open in new window
- Pin to Quick access
- Give access to
- Restore previous versions
- Include in library
- Pin to Start
- Send to
- Cut
- Copy
- Create shortcut
- Delete
- Rename
- Properties

LABFILES				
Home Share View				
This PC > Local Disk (C:) > LABFILES >				
	Name	Date modified	Type	Size
sk access	contains-nothing	10/22/2018 7:57 PM	File folder	
sktop	MARKETING	6/5/2023 1:48 AM	File folder	
wnloads	NVD-Control-RA-5-VULNERABILITY SCA...	6/5/2023 1:48 AM	File folder	
cuments	pcaps	6/5/2023 1:48 AM	File folder	
BFILES	ScoutSuite	6/5/2023 1:48 AM	File folder	
TUP	winlogbeat	6/5/2023 1:48 AM	File folder	
SINTERNALS	515tech_store.sql	3/11/2021 11:13 AM	SQL File	8 KB
PC	comptia-logo	8/22/2018 7:44 AM	JPG File	53 KB
	CONFIDENTIAL	3/28/2019 6:31 AM	Text Document	1 KB
work	conn-sample	3/9/2020 7:00 AM	Text Document	155 KB
	CSIRT Incident Handling Form	2/1/2020 1:37 PM	Microsoft Word D...	16 KB
	CSIRT Incident Handling Form	2/1/2020 1:39 PM	Microsoft Edge P...	104 KB
	DisableController	3/22/2021 2:30 PM	Windows PowerS...	1 KB
	DisableDisk	3/22/2021 11:40 AM	Windows PowerS...	1 KB
	iam_shares	6/28/2022 12:58 AM	Windows PowerS...	2 KB
	laptop-full	2/21/2023 3:44 AM	Wireshark capture...	321 KB
	laptop-selected	2/21/2023 3:47 AM	Wireshark capture...	306 KB
	local.rules	3/10/2020 5:26 AM	RULES File	1 KB
	NVD-Control-RA-5-VULNERABILITY SCA...	1/21/2020 4:29 AM	Microsoft Edge H...	71 KB
	set_default_password	3/11/2021 1:32 AM	Windows PowerS...	1 KB
	Structureality-netdiag	6/28/2022 1:08 AM	OpenDocument D...	44 KB
	trusted-installs	6/29/2022 3:56 AM	OpenOffice.org 1...	1 KB

Issue Found

Folder deletion activity was not being recorded in the Windows Security log because object access auditing was not fully configured.

Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
 - Subscriptions

Security Number of events: 1,207

Keywords	Date a...	Source	Event...	Task Category
Audit ...	6/12/2...	Microsoft Wi...	5379	User Account ...
Audit ...	6/12/2...	Microsoft Wi...	5379	User Account ...
Audit ...	6/12/2...	Microsoft Wi...	5379	User Account ...
Audit ...	6/12/2...	Microsoft Wi...	4672	Special Logon
Audit ...	6/12/2...	Microsoft Wi...	4624	Logon
Audit ...	6/12/2...	Microsoft Wi...	4634	Logoff

Event 5379, Microsoft Windows security auditing.

General Details

Credential Manager credentials were read.

Subject:

Security ID:	structureality\Jaime
Account Name:	Jaime
Account Domain:	structureality
Logon ID:	0x9C472
Read Operation:	Enumerate Credentials

This event occurs when a user performs a read operation on stored credentials in Credential Manager.

Log Name: Security

Source: Microsoft Windows security

Event ID: 5379

Level: Information

User: N/A

OpCode: Info

More Information: [Event Log Online Help](#)

Logged: 6/12/2025 5:59:22 PM

Task Category: User Account Management

Keywords: Audit Success

Computer: PC10.ad.structureality.com

Actions

Security

- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task To this L...
- View
 - Refresh
 - Help
- Event 5379, Microsoft Wind...
 - Event Properties
 - Attach Task To This Ev...
 - Copy
 - Save Selected Events...
 - Refresh
 - Help

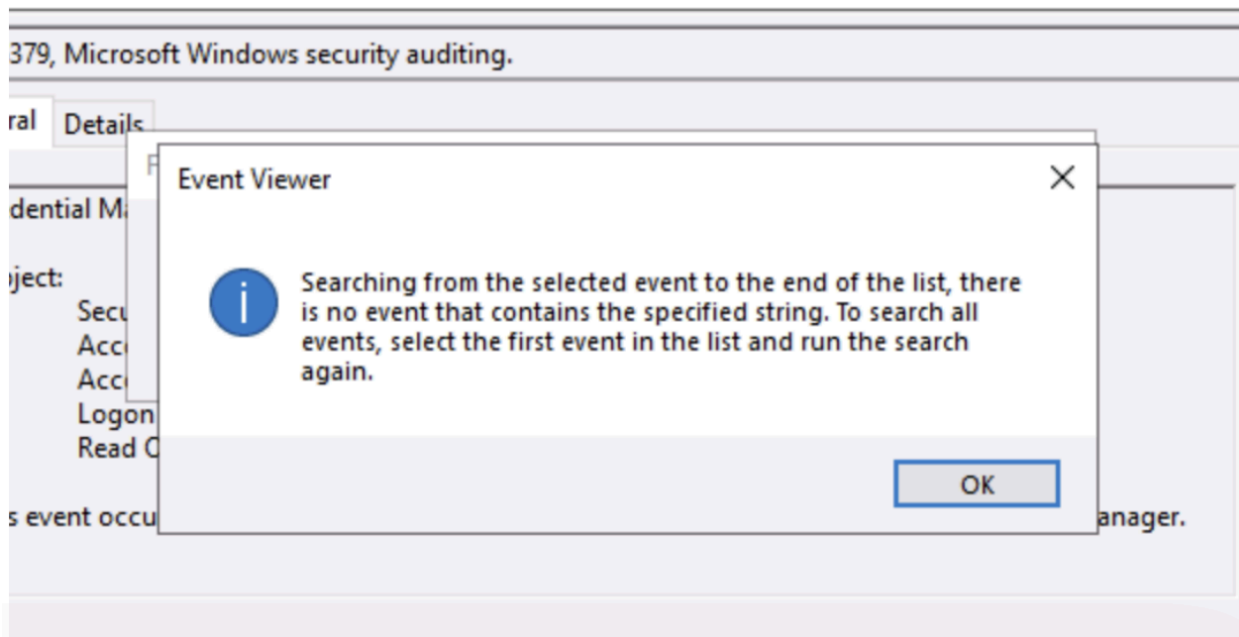
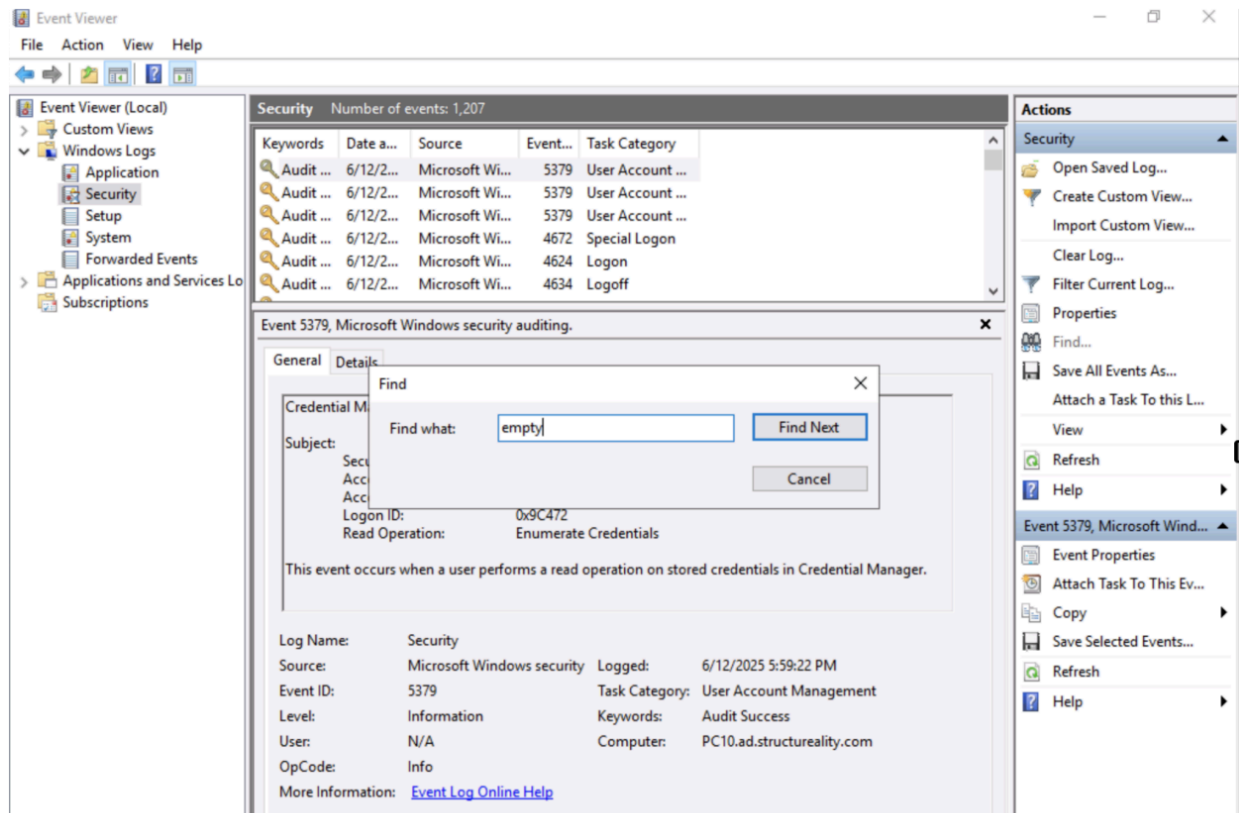


Figure 4: Event Viewer showed that folder deletion was not being audited before configuration.

Action Taken

To enable auditing, I:

1. Opened **Local Security Policy**.
2. Expanded **Local Policies**.
3. Selected **Audit Policy**.
4. Opened **Audit object access**.
5. Enabled both **Success** and **Failure** auditing.
6. Opened the properties of the **LABFILES** folder.
7. Went to **Advanced Security Settings**.
8. Added an auditing entry for **Everyone**.
9. Enabled auditing for **Delete subfolders and files** and **Delete**.
10. Deleted the **pcaps** folder.
11. Opened Event Viewer.
12. Searched for Event ID **4660** and **4663**.

Event ID **4660** showed that an object was deleted, while Event ID **4663** showed the object name, confirming that the deleted folder was **C:\LABFILES\pcaps**.



Local Security Policy

Desktop app

Apps

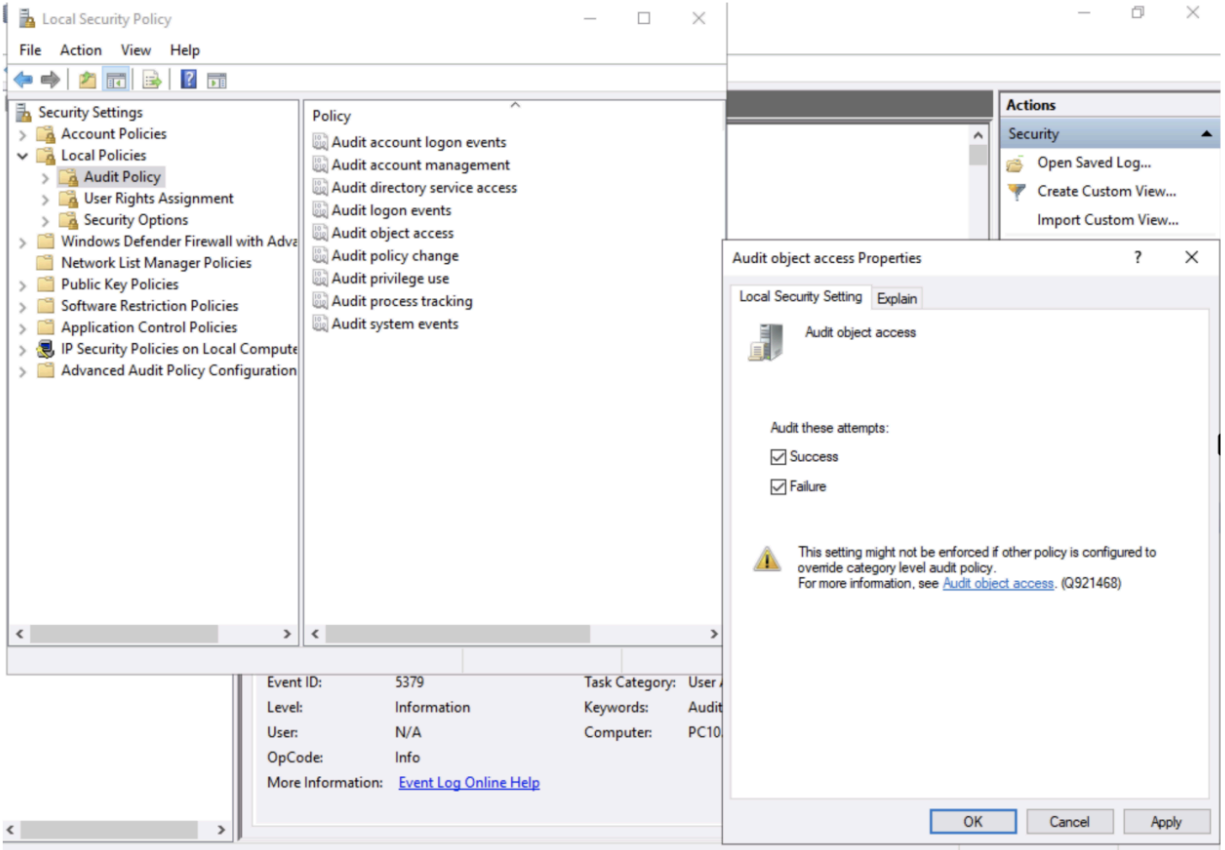
 This PC

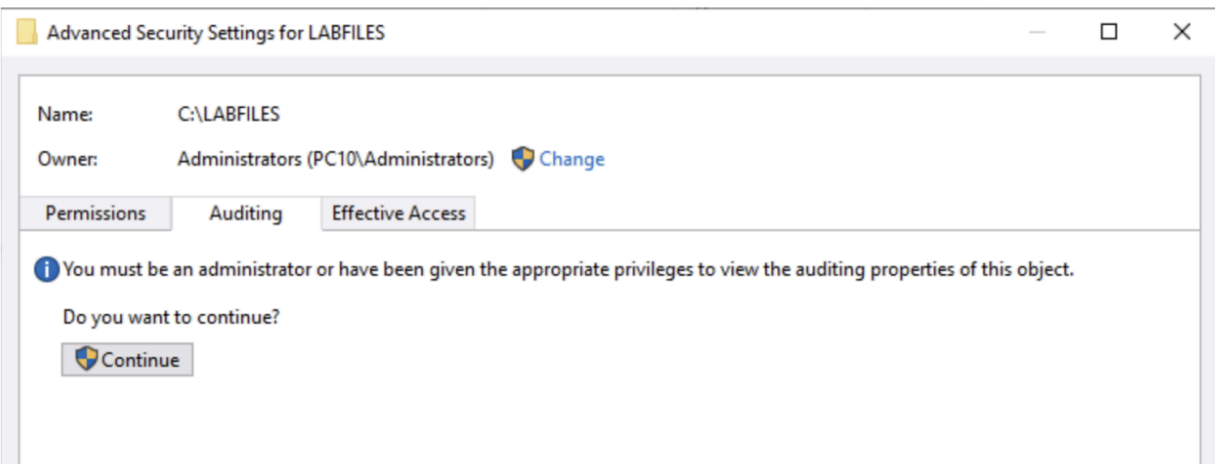
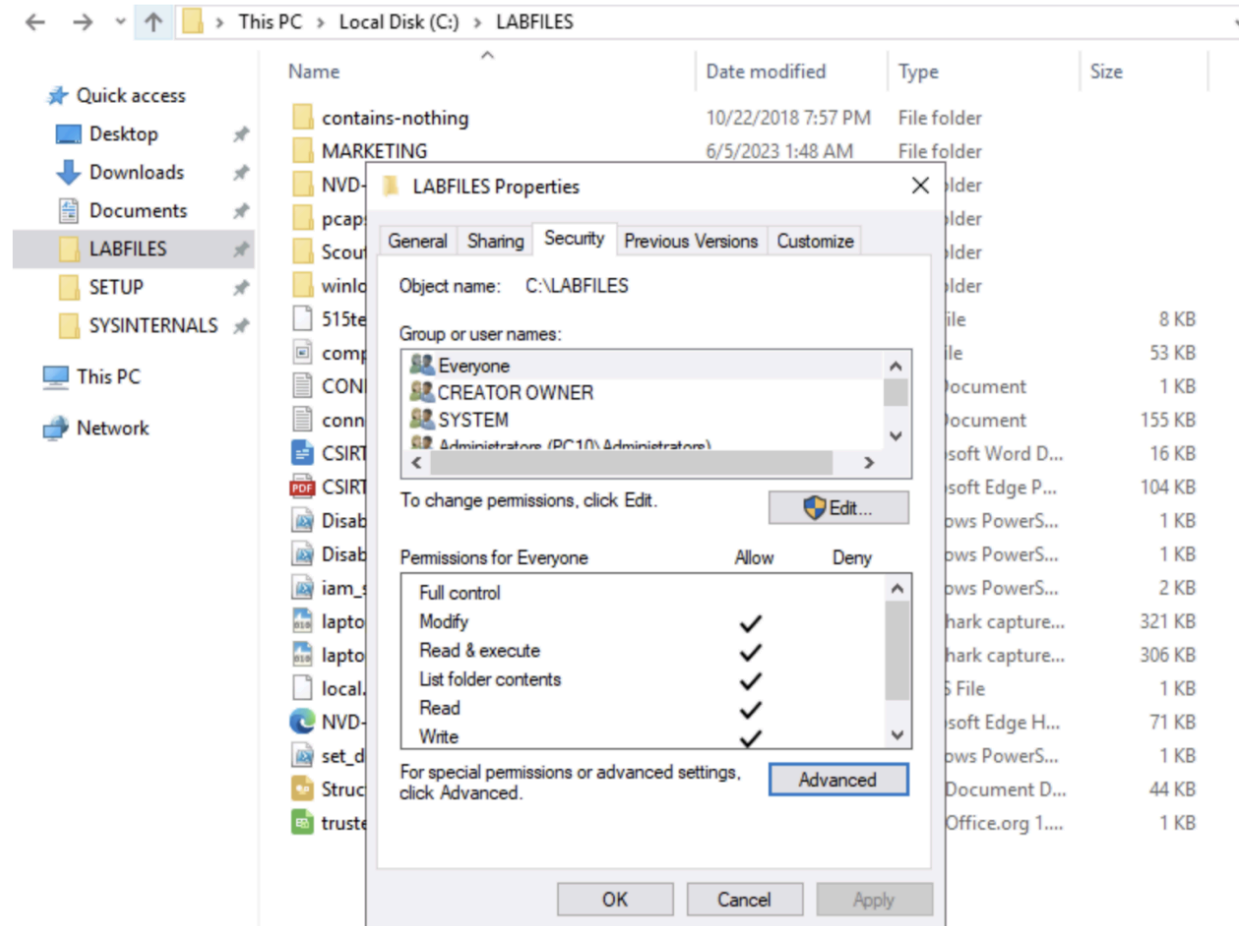
Settings

-  Language settings
-  Edit language and keyboard options
-  Choose if websites can use your language list
-  Add a language to this device
-  Change the language that Windows features appear in
-  Choose the language you speak with your device
-  Recognize non-native accents

 local|Security Policy







Advanced

Auditing Entry for LABFILES

Name:

Owner:

Permission:

Principal: [Select a principal](#)

Type:

Success

Applies to:

This folder, subfolders and files

For additional auditing entries, click Add.

Auditing entry type

Type

Add

Enable inheritance

☐ Replace existing permissions

Basic permissions:

☐ Full control

☐ Modify

☒ Read & execute

☒ List folder contents

☒ Read

☐ Write

☐ Special permissions

☐ Only apply these auditing settings to objects and/or containers within this container

Show advanced permissions

Add a condition to limit the scope of this auditing entry. Security events will be logged only if conditions are met.

Add a condition

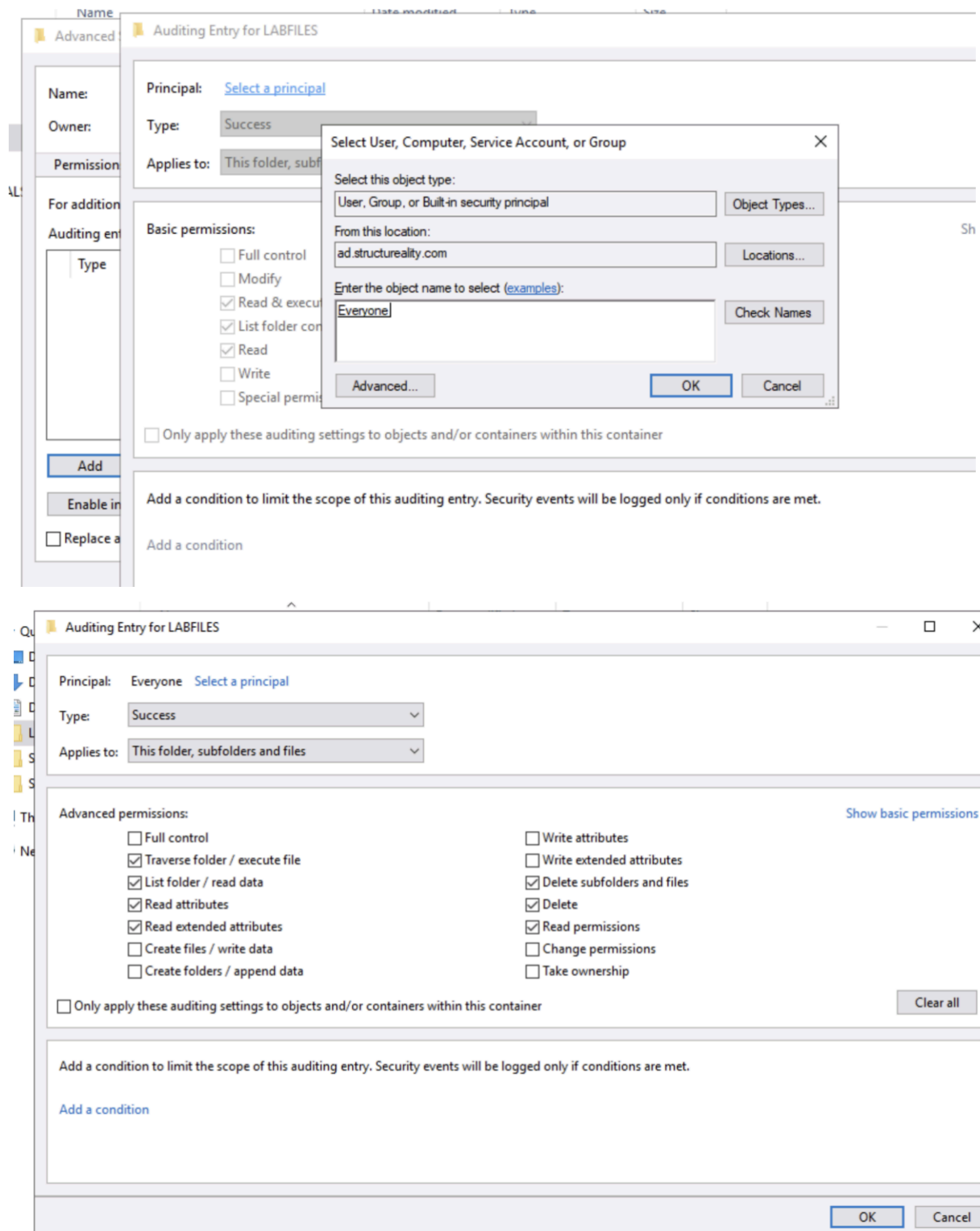
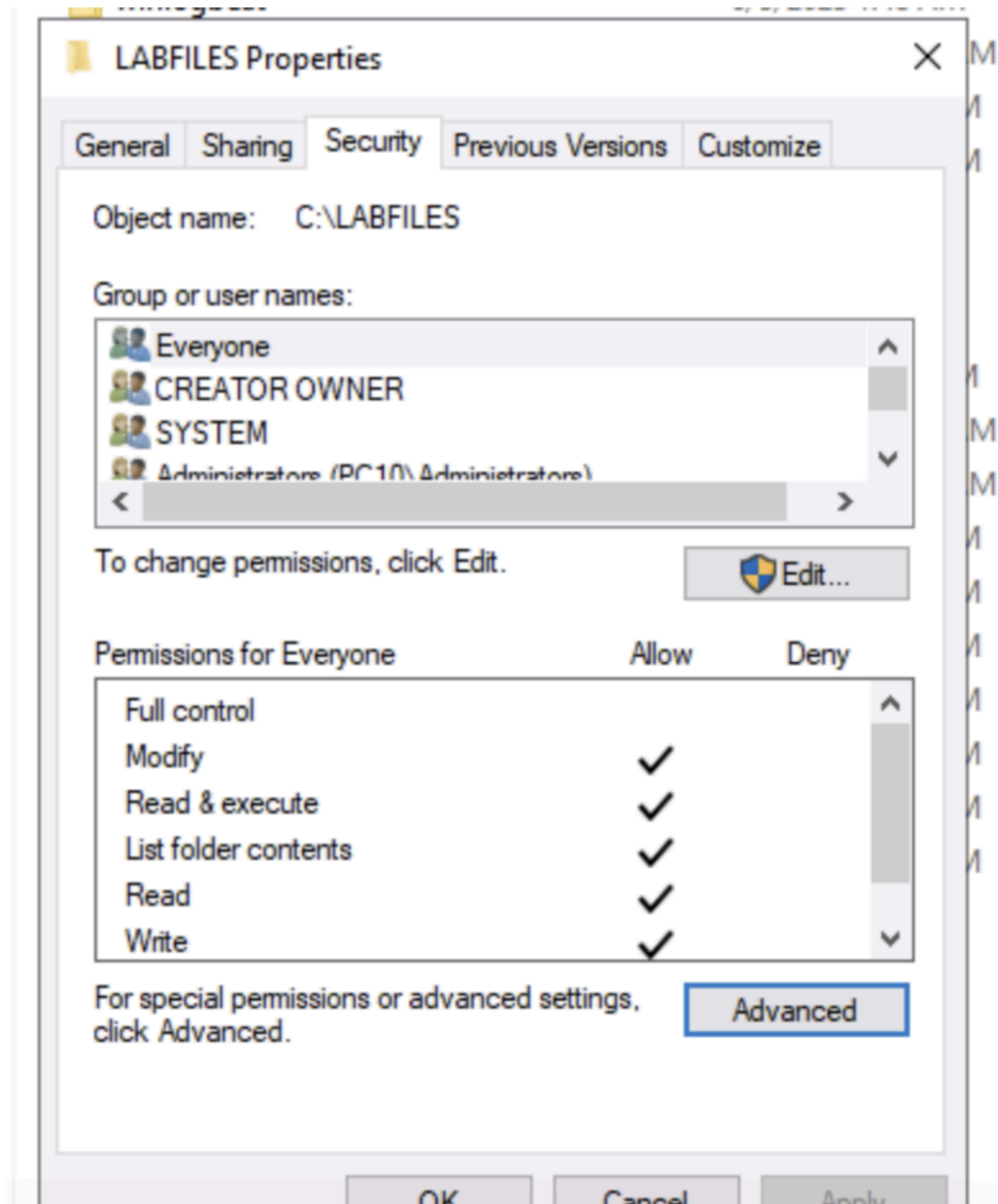


Figure 5: Auditing was configured on the LABFILES folder to record delete activity.

Result

The detective control successfully created a record of the folder deletion activity in Event Viewer.



File Home Share View

← → ↕ ↑ > This PC > Local Disk (C:) > LABFILES

Name	Date modified	Type	Size
contains-nothing	10/22/2018 7:57 PM	File folder	
MARKETING	6/5/2023 1:48 AM	File folder	
NVD-Control-RA-5-VULNERABILITY SCA...	6/5/2023 1:48 AM	File folder	
pcaps	6/5/2023 1:48 AM	File folder	
ScoutSuite	6/5/2023 1:48 AM	File folder	
winlogbeat	6/5/2023 1:48 AM	File folder	
515tech_store.sql	3/11/2021 11:13 AM	SQL File	8 KB
comptia-logo	8/22/2018 7:44 AM	JPG File	53 KB
CONFIDENTIAL	3/28/2019 6:31 AM	Text Document	1 KB
conn-sample	3/9/2020 7:00 AM	Text Document	155 KB
CSIRT Incident Handling Form	2/1/2020 1:37 PM	Microsoft Word D...	16 KB
CSIRT Incident Handling Form	2/1/2020 1:39 PM	Microsoft Edge P...	104 KB
DisableController	3/22/2021 2:30 PM	Windows PowerS...	1 KB
DisableDisk	3/22/2021 11:40 AM	Windows PowerS...	1 KB
iam_shares	6/28/2022 12:58 AM	Windows PowerS...	2 KB
laptop-full	2/21/2023 3:44 AM	Wireshark capture...	321 KB
laptop	2/23/2023 3:47 AM	Wireshark capture...	306 KB
local	10/20/2020 5:26 AM	RULES File	1 KB
NVD-Control-RA-5-VULNERABILITY SCA...	10/20/2020 4:29 AM	Microsoft Edge H...	71 KB
set_d	10/21/2021 1:32 AM	Windows PowerS...	1 KB
Structure...	10/22/2022 1:08 AM	OpenDocument D...	44 KB

Quick access

- Desktop
- Downloads
- Documents
- LABFILES
- SETUP
- SYSINTERNALS
- This PC
- Network

Open

- Open in new window
- Pin to Quick access
- Give access to
- Restore previous versions
- Include in library
- Pin to Start
- Send to
- Cut
- Copy
- Create shortcut
- Delete
- Rename
- Properties

This PC > Local Disk (C:) > LABFILES >

Name	Date modified	Type	Size
contains-nothing	10/22/2018 7:57 PM	File folder	
MARKETING	6/5/2023 1:48 AM	File folder	
NVD-Control-RA-5-VULNERABILITY SCA...	6/5/2023 1:48 AM	File folder	
ScoutSuite	6/5/2023 1:48 AM	File folder	
winlogbeat	6/5/2023 1:48 AM	File folder	
515tech_store.sql	3/11/2021 11:13 AM	SQL File	8 KB
comptia-logo	8/22/2018 7:44 AM	JPG File	53 KB
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CSIRT Incident Handling Form	2/1/2020 1:39 PM	Microsoft Edge P...	104 KB
DisableController	3/22/2021 2:30 PM	Windows PowerS...	1 KB
DisableDisk	3/22/2021 11:40 AM	Windows PowerS...	1 KB
iam_shares	6/28/2022 12:58 AM	Windows PowerS...	2 KB
laptop-full	2/21/2023 3:44 AM	Wireshark capture...	321 KB

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
 - Subscriptions

Security Number of events: 2,260 (!) New events available

Keywords	Date a...	Source	Event...	Task Category
Audit ...	6/12/2...	Microsoft Wi...	4662	Other Object A...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...

Event 4662, Microsoft Windows security auditing.

General Details

An operation was performed on an object.

Subject:

- Security ID: LOCAL SERVICE
- Account Name: LOCAL SERVICE
- Account Domain: NT AUTHORITY
- Logon ID: 0x3E5

Object:

- Object Server: LSA
- Object Type: SecretObject
- Object Name: Policy\Secrets\SMACHINE.ACC

Log Name: Security

Source: Microsoft Windows security Logged: 6/12/2025 6:56:07 PM

Event ID: 4662 Task Category: Other Object Access Events

Level: Information Keywords: Audit Success

User: N/A Computer: PC10.ad.structureality.com

OpCode: Info

More Information: [Event Log Online Help](#)

Actions

- Security
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task To this L...
- View
- Refresh
- Help
- Event 4662, Microsoft Wind...
- Event Properties
- Attach Task To This Ev...
- Copy
- Save Selected Events...
- Refresh
- Help

File Action View Help

Event Viewer (Local)

- Custom Views
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Audit ...	6/12/2...	Microsoft Wi...	4662	Other Object A...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...
Audit ...	6/12/2...	Microsoft Wi...	5152	Filtering Platfo...

Event 4662, Microsoft Windows security auditing.

General Details

Security

Number of events: 2,260 (!) New events available

Keywords	Date a...	Source	Event...	Task Category
Audit ...	6/12/2...	Microsoft Wi...	4660	File System
Audit ...	6/12/2...	Microsoft Wi...	4663	File System
Audit ...	6/12/2...	Microsoft Wi...	4663	File System
Audit ...	6/12/2...	Microsoft Wi...	4656	File System
Audit ...	6/12/2...	Microsoft Wi...	4658	File System
Audit ...	6/12/2...	Microsoft Wi...	4690	Handle Manip...

Event 4660, Microsoft Windows security auditing.

General

Details

An object wa

Subject:

Secu

Acc

Acc

Login ID: 0xTA2CDE

Object:

Object Server: Security

Actions

Security

Open Saved Log...

Create Custom View...

Import Custom View...

Clear Log...

Filter Current Log...

Properties

Find...

Save All Events As...

Attach a Task To this L...

View

Refresh

Help

Event 4660, Microsoft Wind...

Event Properties

Find

Find what: 4660

Find Next

Cancel

Security

Number of events: 2,260 (!) New events available

Keywords	Date a...	Source	Event...	Task Category
Audit ...	6/12/2...	Microsoft Wi...	4660	File System
Audit ...	6/12/2...	Microsoft Wi...	4663	File System
Audit ...	6/12/2...	Microsoft Wi...	4663	File System
Audit ...	6/12/2...	Microsoft Wi...	4656	File System
Audit ...	6/12/2...	Microsoft Wi...	4658	File System
Audit ...	6/12/2...	Microsoft Wi...	4690	Handle Manip...

Event 4660, Microsoft Windows security auditing.

General

Details

An object was deleted.

Subject:

Security ID:

structureality\Jaime

Account Name:

Jaime

Account Domain:

structureality

Logon ID:

0x1A2CDE

Object:

Object Server:

Security

Handle ID:

0x618

Log Name:

Security

Source:

Microsoft Windows security

Logged:

6/12/2025 6:53:50 PM

Event ID:

4660

Task Category:

File System

Level:

Information

Keywords:

Audit Success

User:

N/A

Computer:

PC10.ad.structureality.com

OpCode:

Info

More Information:

[Event Log Online Help](#)

Event Properties - Event 4660, Microsoft Windows security auditing.

General Details

An object was deleted.

Subject:

Security ID: structureality\Jaime
Account Name: Jaime
Account Domain: structureality
Logon ID: 0x3D3A9

Log Name: Security

Source: Microsoft Windows security Logged: 7/15/2025 10:45:10 AM

Event ID: 4660 Task Category: File System

Level: Information Keywords: Audit Success

User: N/A Computer: PC10.ad.structureality.com

OpCode: Info

More Information: [Event Log Online Help](#)

Copy

Clos

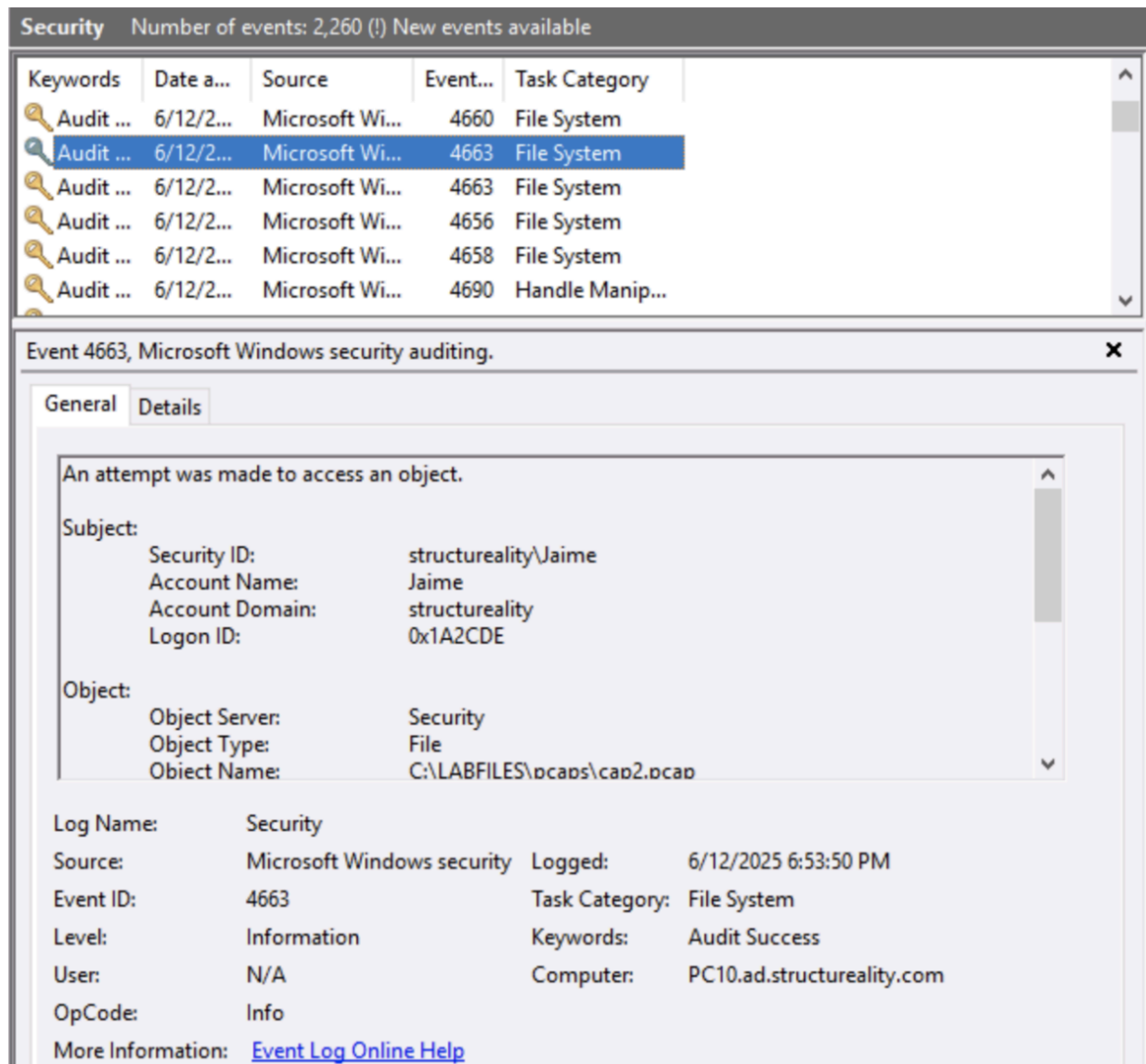


Figure 6: Event ID 4663 confirmed the deleted object was C:\LABFILES\pcaps.

Part 3: Configured and Tested Directive Controls

What I Did

The third part of the lab focused on a **directive control**. A directive control gives instructions to users and guides them toward compliant behavior.

In this lab, I created a login warning banner using PowerShell. The banner informed users that the system was for authorized use only and that use of the system required compliance with the Acceptable Use Policy.

Commands Used

\$BannerText = "This computer system is the property of Structureality Inc. It is for authorized use only. By using this system, all users acknowledge notice of and agree to comply with the Acceptable Use Policy (AUP). Unauthorized or improper use of this system may result in administrative disciplinary action, civil charges/criminal penalties, and/or other sanctions set forth in the AUP. By continuing to use this system, we indicate our awareness of and consent to these terms and conditions. If we are physically located in the European Union, we may have additional rights per the GDPR. Visit the website gdpr-info.eu for more information."

```
New-ItemProperty -Path "HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System"
-Name "legalnoticecaption" -Value "Authorized Use Only" -PropertyType "String" -Force |
Out-Null
```

```
New-ItemProperty -Path "HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System"
-Name "legalnoticetext" -Value $BannerText -PropertyType "String" -Force | Out-Null
```



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\system32> $BannerText = "This computer system is the property of Structureality Inc. It is for authorized
use only. By using this system, all users acknowledge notice of and agree to comply with the Acceptable Use Policy (AUP)
. Unauthorized or improper use of this system may result in administrative disciplinary action, civil charges/criminal p
enalties, and/or other sanctions set forth in the AUP. By continuing to use this system, you indicate your awareness of
and consent to these terms and conditions. If you are physically located in the European Union, you may have additional
rights per the GDPR. Visit the website gdpr-info.eu for more information."
PS C:\Windows\system32> New-ItemProperty -Path "HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System" -Name "
legalnoticecaption" -Value "Authorized Use Only" -PropertyType "String" -Force | Out-Null
PS C:\Windows\system32> New-ItemProperty -Path "HKLM:\Software\Microsoft\Windows\CurrentVersion\Policies\System" -Name "
legalnoticetext" -Value $BannerText -PropertyType "String" -Force | Out-Null
PS C:\Windows\system32> _
```

Result

After signing out and sending **Ctrl + Alt + Delete**, the login warning banner appeared. This confirmed that the directive control was successfully implemented. The goal of directive controls is compliance.

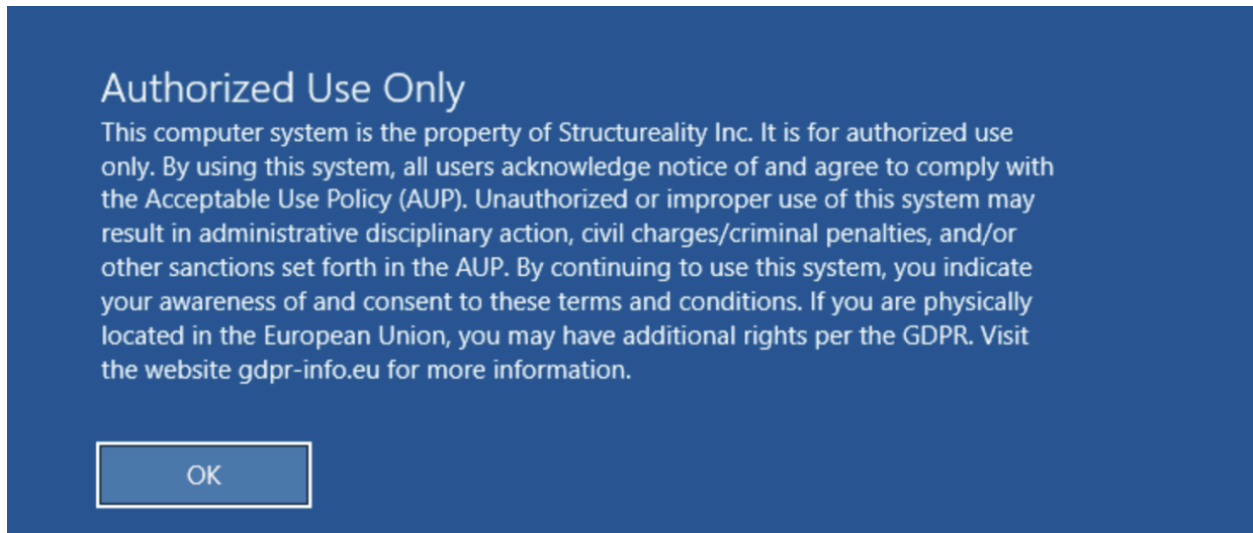


Figure 7: Login warning banner displayed after configuring the directive control.

Part 4: Configured and Tested Corrective Controls

What I Did

The final part of the lab focused on a **corrective control**. A corrective control helps return a system to a normal or secure state after an unwanted event occurs.

First, I used the Sysinternals **NotMyFault** tool to trigger a Windows stop error/BSOD. This demonstrated that Windows has native corrective protections that stop misbehaving applications and reboot the system to protect the execution environment.

File Explorer window showing the contents of the SYSINTERNALS folder on the Local Disk (C:).

Navigation pane (Left):

- Quick access
- Desktop
- Downloads
- Documents
- LABFILES
- SETUP
- SYSINTERNALS (selected)
- This PC
- Network

Address bar: This PC > Local Disk (C:) > SYSINTERNALS

Search bar: Search SYSINTERNALS

Name	Date modified	Type	Size
logonsessions04	6/28/2022 12:55 AM	Application	330 KB
movefile	6/28/2022 12:55 AM	Application	331 KB
movefile64	6/28/2022 12:55 AM	Application	430 KB
notmyfault	6/28/2022 12:55 AM	Application	505 KB
notmyfault64	6/28/2022 12:55 AM	Application	637 KB
notmyfaultc	6/28/2022 12:55 AM	Application	503 KB
notmyfaultc64	6/28/2022 12:55 AM	Application	635 KB
ntfsinfo	6/28/2022 12:55 AM	Application	137 KB
ntfsinfo64	6/28/2022 12:55 AM	Application	156 KB
pendmoves	6/28/2022 12:55 AM	Application	333 KB
pendmoves64	6/28/2022 12:55 AM	Application	431 KB
pipelist	6/28/2022 12:55 AM	Application	332 KB
pipelist64	6/28/2022 12:55 AM	Application	432 KB
portmon	6/28/2022 12:55 AM	Application	441 KB
procdump	6/28/2022 12:55 AM	Application	736 KB
procdump64	6/28/2022 12:55 AM	Application	392 KB
procexp	6/28/2022 12:55 AM	Compiled HTML ...	71 KB
procexp	6/28/2022 12:55 AM	Application	2,773 KB
procexp64	6/28/2022 12:55 AM	Application	1,475 KB
procmon	6/28/2022 12:55 AM	Compiled HTML ...	63 KB
Procmon	6/28/2022 12:55 AM	Application	5,091 KB
Procmon64	6/28/2022 12:55 AM	Application	2,631 KB

160 items | 1 item selected 634 KB

Not My Fault



Crash

Hang

Leak

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Contributions by Daniel Pearson

Options

Clicking the Crash button will cause the system to crash.
There is a risk that corrupted memory will be written to disk
or that work may be lost. Close any open applications.

- ☐ High IRQL fault (Kernel-mode)
- ☐ Buffer overflow
- ☒ Code overwrite
- ☐ Stack trash
- ☐ High IRQL fault (User-mode)
- ☐ Stack overflow
- ☐ Hardcoded breakpoint
- ☐ Double free

Colors...

Crash

Cancel

Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

50% complete



For more information about this issue and possible fixes, visit <https://www.windows.com/stopcode>

If you call a support person, give them this info:

Stop code: IRQL_NOT_LESS_OR_EQUAL

Next, I created a custom corrective control using PowerShell. The goal was to monitor a file named `notes.txt`, detect if its content changed, and restore the file to the correct content.

PowerShell Commands Used

Created the file:

```
"This is important" | Set-Content notes.txt
```

Displayed the file:

```
type notes.txt
```

Generated a SHA256 hash:

```
Get-FileHash ./notes.txt -Algorithm SHA256 | Select-Object -ExpandProperty Hash |  
Set-Content ./hash.txt
```

Modified the file:

```
echo blah >> notes.txt
```

Checked whether the file changed:

```
if((Get-FileHash ./notes.txt -Algorithm SHA256).Hash -eq (Get-Content ./hash.txt)) {  
    Write-Host "The file is correct."  
} else {  
    Write-Host "The file has changed. Corrective action should be initiated."  
}
```

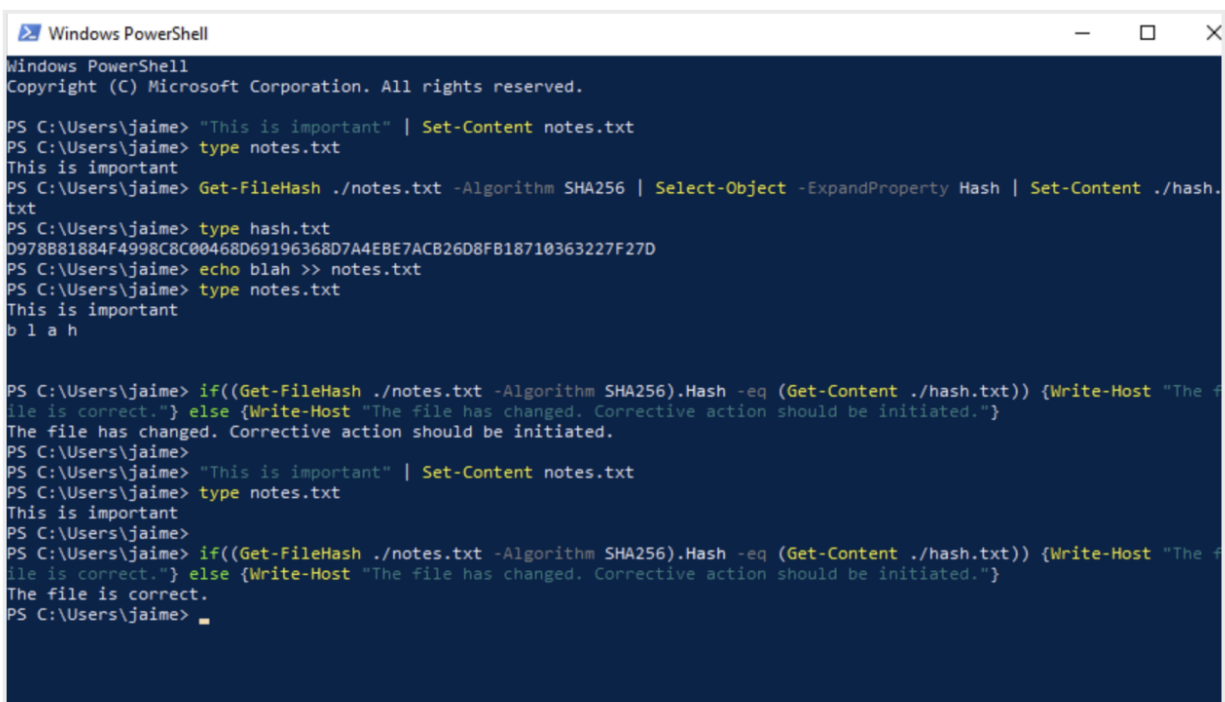
Restored the file:

"This is important" | Set-Content notes.txt

Created a script named `check.ps1`:

```
if((Get-FileHash ./notes.txt -Algorithm SHA256).Hash -ne (Get-Content ./hash.txt))
{
    "This is important" | Set-Content ./notes.txt
    Write-Host "The file has changed. Corrective action initiated."
}
else
{
    Write-Host "The file is correct. No corrective action needed."
}
```

The lab confirmed that if the file changed, the PowerShell script restored the correct content. This simulated a corrective control similar to Windows Signature Verification behavior.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\jaime> "This is important" | Set-Content notes.txt
PS C:\Users\jaime> type notes.txt
This is important
PS C:\Users\jaime> Get-FileHash ./notes.txt -Algorithm SHA256 | Select-Object -ExpandProperty Hash | Set-Content ./hash.
txt
PS C:\Users\jaime> type hash.txt
D978B81884F4998C8C00468D69196368D7A4EBE7ACB26D8F818710363227F27D
PS C:\Users\jaime> echo blah >> notes.txt
PS C:\Users\jaime> type notes.txt
This is important
b l a h

PS C:\Users\jaime> if((Get-FileHash ./notes.txt -Algorithm SHA256).Hash -eq (Get-Content ./hash.txt)) {Write-Host "The f
ile is correct."} else {Write-Host "The file has changed. Corrective action should be initiated."}
The file has changed. Corrective action should be initiated.
PS C:\Users\jaime>
PS C:\Users\jaime> "This is important" | Set-Content notes.txt
PS C:\Users\jaime> type notes.txt
This is important
PS C:\Users\jaime>
PS C:\Users\jaime> if((Get-FileHash ./notes.txt -Algorithm SHA256).Hash -eq (Get-Content ./hash.txt)) {Write-Host "The f
ile is correct."} else {Write-Host "The file has changed. Corrective action should be initiated."}
The file is correct.
PS C:\Users\jaime> █
```

Result

The corrective control successfully detected unauthorized file changes and restored the file to the approved content.


```

PS C:\Users\jaime>
PS C:\Users\jaime> if((Get-FileHash ./notes.txt -Algorithm SHA256).Hash -eq (Get-Content ./hash.txt)) {Write-Host "The
file is correct."} else {Write-Host "The file has changed. Corrective action should be initiated."}
The file is correct.
PS C:\Users\jaime> notepad calchash.ps1
PS C:\Users\jaime> ./calchash.ps1
PS C:\Users\jaime> type hash.txt
D978B81884F4998C8C00468D69196368D7A4EBE7ACB26D8FB18710363227F27D
PS C:\Users\jaime> notepad check.ps1
PS C:\Users\jaime> ./check.ps1
The file is correct. No corrective action needed.
PS C:\Users\jaime>

```

```

The file is correct. } else {Write-Host "The file has changed. Corrective action
The file is correct.
PS C:\Users\jaime> notepad calchash.ps1
PS C:\Users\jaime> ./calchash.ps1
PS C:\Users\jaime> type hash.txt
D978B81884F4998C8C00468D69196368D7A4EBE7ACB26D8FB18710363227F27D
PS C:\Users\jaime> notepad check.ps1
PS C:\Users\jaime> ./check.ps1
The file is correct. No corrective action needed.
PS C:\Users\jaime> type notes.txt
This is important
PS C:\Users\jaime> echo blah >> notes.txt
PS C:\Users\jaime> type notes.txt
This is important
b l a h

PS C:\Users\jaime> ./check.ps1
The file has changed. Corrective action initiated.
PS C:\Users\jaime> type notes.txt
This is important
PS C:\Users\jaime>

```

Figure 8: PowerShell script detected a file change and restored notes.txt to the approved content.

8. Key Findings

Finding	Explanation
Non-admin users had improper access	Sam could access the TOOLS share because the Users group inherited read permissions.
Preventive control fixed the access issue	Removing unnecessary inherited permissions blocked Sam from accessing the share.
Folder deletion was not audited by default	Event Viewer did not show the deleted empty folder until auditing was enabled.

Detective control created useful logs	Event IDs 4660 and 4663 confirmed object deletion activity.
Login banners support compliance	The directive control informed users about authorized system use.
PowerShell can support corrective controls	Hash comparison detected file changes and restored the correct file content.

9. Analysis

This lab showed how each type of security control plays a different role in protecting systems.

The **preventive control** blocked unauthorized access before it could continue. The **detective control** created records that could help investigate file deletion activity. The **directive control** communicated rules and expectations to users before login. The **corrective control** detected file changes and restored the file to its approved state.

This is important because real-world security does not depend on only one control. A strong security posture uses multiple layers of controls to prevent, detect, guide, and correct security-related events.

10. Recommendation

Based on this lab, I would recommend the following:

1. Review shared folder permissions regularly.
2. Remove unnecessary inherited permissions from sensitive folders.
3. Follow the principle of least privilege.
4. Enable auditing for important folders and security-sensitive actions.
5. Monitor Event Viewer logs for suspicious activity.
6. Use login banners when required by company policy or compliance standards.
7. Use file integrity monitoring for critical files.
8. Automate corrective scripts when appropriate.
9. Document permission changes and security findings.
10. Escalate violations to the security team when needed.

11. IT Support Relevance

This lab is highly relevant to IT Support Analyst roles because support analysts often help with access issues, folder permissions, login policies, user activity logs, and basic security troubleshooting.

This lab connects to real IT support tasks such as:

- Troubleshooting “access denied” issues
- Reviewing file and folder permissions
- Understanding group-based access
- Supporting Windows auditing
- Checking Event Viewer logs
- Explaining login banners and acceptable use policies
- Using PowerShell for basic automation
- Understanding file integrity and hashing
- Documenting technical changes

For an entry-level IT support role, this lab shows that I understand how security controls work in a Windows environment and how they apply to real user and system issues.

12. Skills Demonstrated

This lab demonstrates the following skills:

- Windows Server 2019 administration
- File share permission management
- NTFS permissions
- Access control troubleshooting
- Principle of least privilege
- Windows auditing
- Event Viewer log analysis
- Local Security Policy configuration
- PowerShell scripting
- Registry-based login banner configuration
- SHA256 hashing
- File integrity monitoring
- Corrective control testing
- Technical documentation
- Security control analysis

13. Ticket-Style Summary

Issue:

The organization needed to test and configure examples of preventive, detective, directive, and corrective security controls.

Impact:

Improper permissions allowed non-administrative users to view restricted resources, and folder deletion activity was not initially audited.

Troubleshooting Steps:

- Tested access to the **TOOLS** share using a non-admin account.
- Identified inherited permissions that allowed unnecessary access.
- Removed improper **Users** and **Everyone** permissions.
- Tested access again to confirm the preventive control worked.
- Deleted a folder and checked Event Viewer for audit logs.
- Enabled object access auditing through Local Security Policy.
- Configured folder-level auditing on **C:\LABFILES**.
- Verified Event IDs 4660 and 4663 after deleting a folder.
- Created a login warning banner using PowerShell.
- Tested the login banner after signing out.
- Used PowerShell hashing to detect changes to **notes.txt**.
- Created a corrective script to restore the file when changed.

Root Cause:

The restricted share had inherited permissions that allowed non-administrative access, and object deletion auditing was not fully configured.

Resolution/Recommendation:

Apply least privilege permissions, enable auditing on sensitive folders, use directive warnings for compliance, and automate corrective actions for critical files when appropriate.

Status:

Completed successfully.

14. Lessons Learned

In this lab, I learned that security controls work together to protect systems in different ways. A preventive control can stop unauthorized access, but detective controls are still needed to record activity. Directive controls help users understand rules and expectations, while corrective controls help return systems or files to the correct state after a problem occurs.

I also learned that Windows permissions can be affected by inheritance, and this can accidentally give users more access than they should have. This is why IT teams must carefully review permissions, auditing settings, and security policies.